

Emertex Smart Edge-AI EV Charging Optimizer

Predictive load management and execution stack
for constrained infrastructure.

Feeder Capacity (64A) < Simultaneous Demand (3x 32A)



Electrical Hazard

Simultaneous full-current charging leads to absolute feeder overload, risking a tripped breaker or electrical fire.



Thermal Hazard

Sustained high-current draw causes connector temperatures to rise rapidly, risking permanent cable damage.



Capacity Gridlock

Without dynamic control, a newly arriving EV encounters zero electrical headroom and cannot charge.

Required: A sub-second control loop to sense, predict, and react to physical constraints autonomously.

OPTIMIZATION STRATEGY: DYNAMIC TRAFFIC-LIGHT GAUGE SYSTEM



ALLOW

Up to 32A allocation.
Safe thermal and
electrical headroom.



THROTTLE

Reduced current allocation.
Dynamic throttling to
manage connector heat
build-up or share limited
feeder capacity.

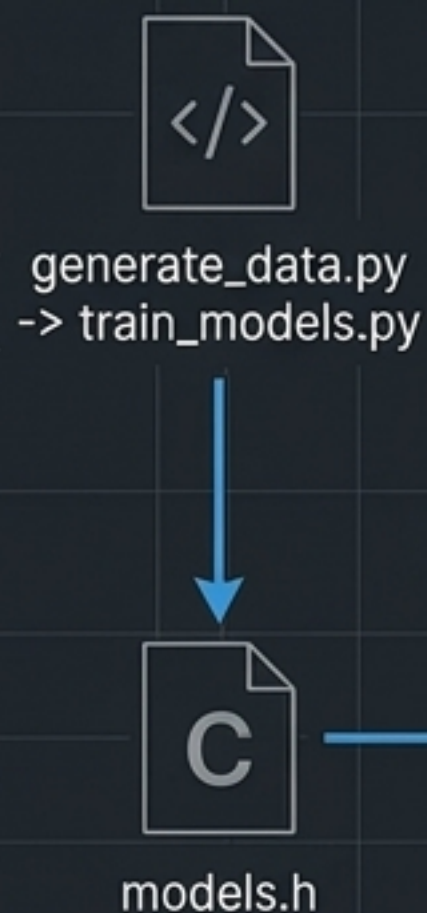


DEFER

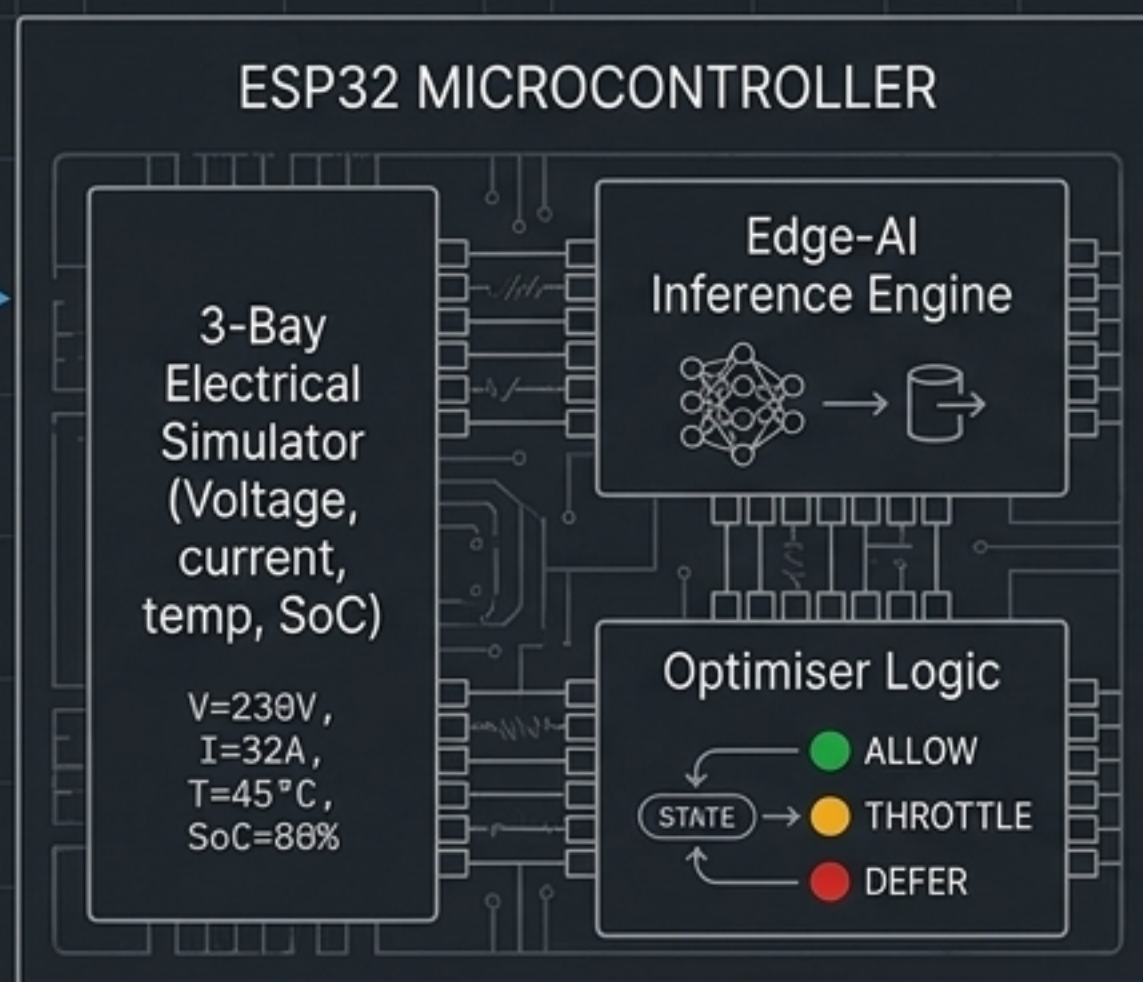
0A paused session. Strictly
enforced to prevent
crossing the 64A feeder
limit or 80°C critical
thermal threshold.

ARCHITECTURAL FLOW: OFFLINE TRAINING TO EDGE EXECUTION TO DUAL-CLOUD TELEMETRY

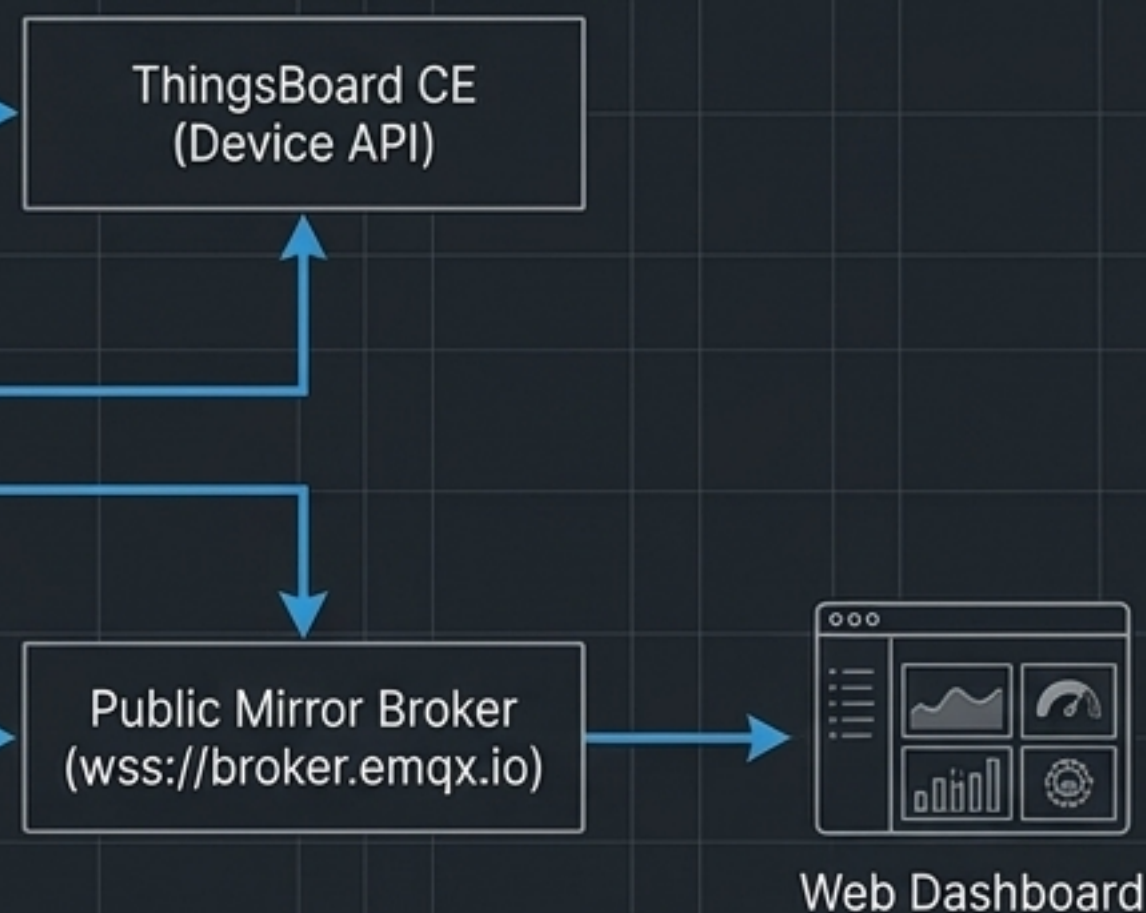
PHASE 1: OFFLINE TRAINING



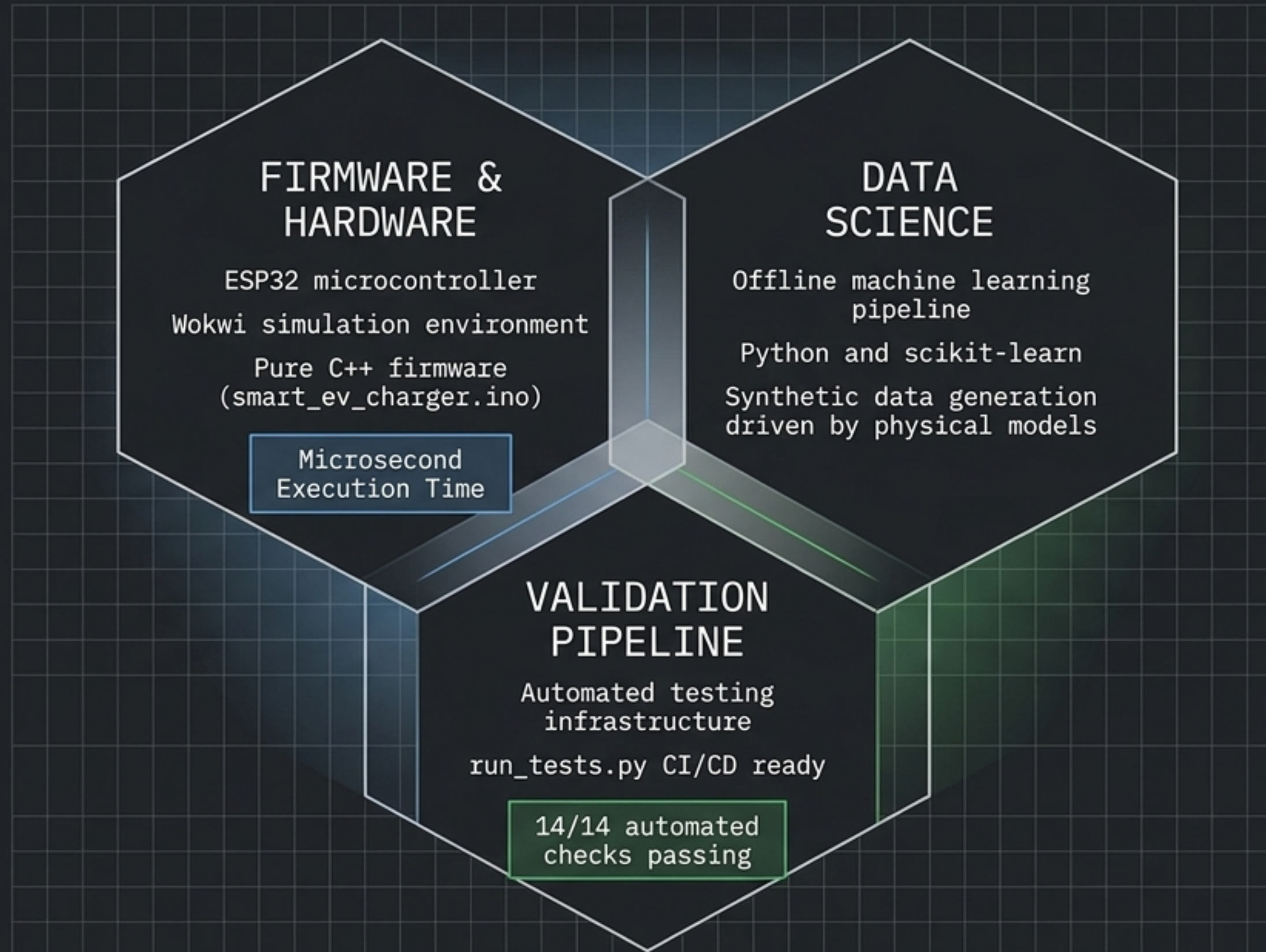
PHASE 2: EDGE EXECUTION



PHASE 3: DUAL-CLOUD TELEMETRY



EXECUTION STACK TECHNOLOGICAL TRIAD



Model A: Arrival Probability

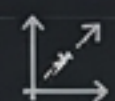
Will an EV plug in during the next 15 minutes?



Algorithm:
Logistic Regression



Scale: Trained on 80,000 synthetic samples



Features: 6 dimensions
(time, busy rate, peak hours)

ROC-AUC: 0.79

Model B: Session Duration

How many minutes will this specific session take?



Algorithm: Decision Tree
Regressor (Max Depth 8, 511 nodes)



Scale: Trained on 60,000 synthetic samples

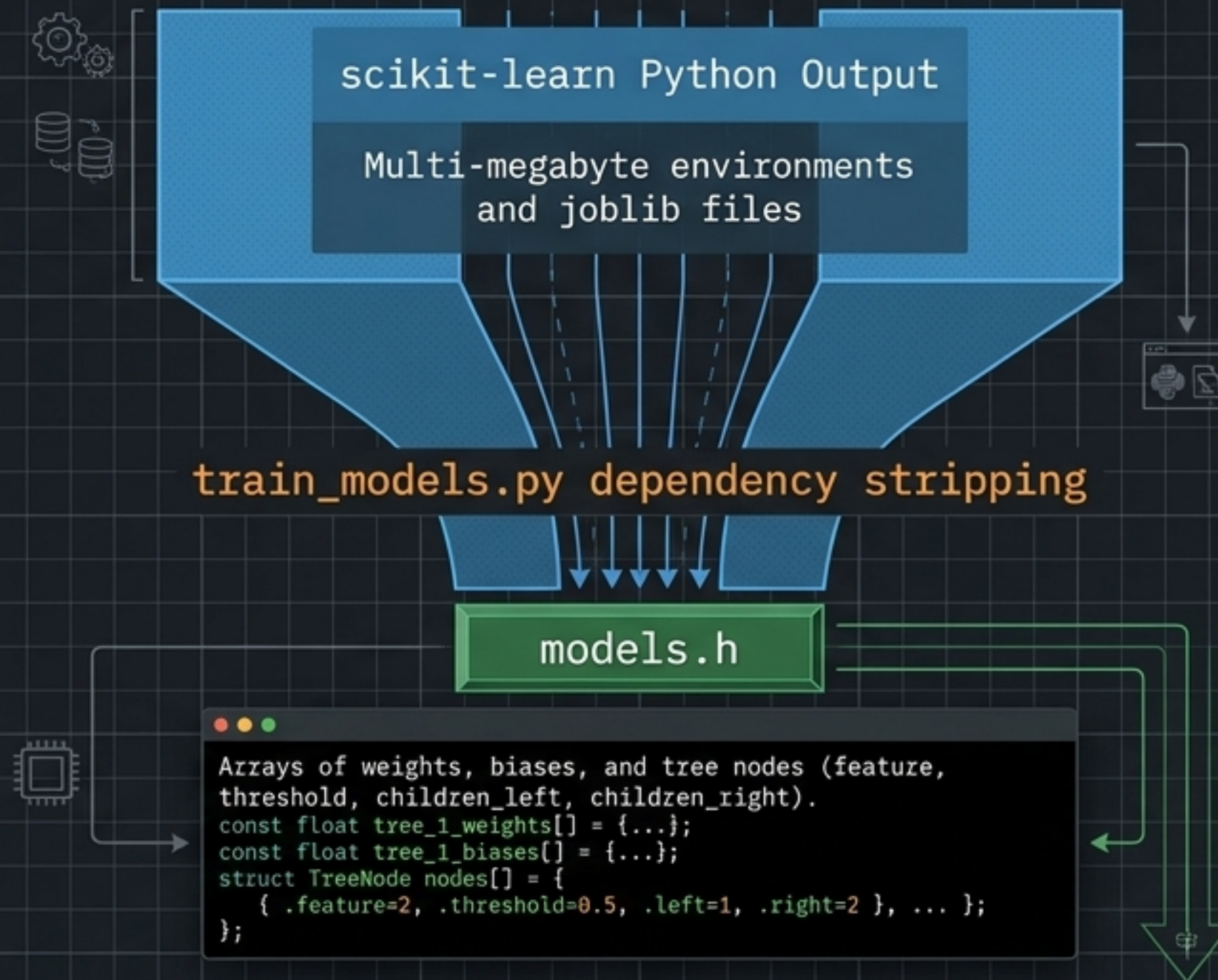


Features: 6 dimensions
(capacity, SoC, temperatures)

MAE: ~43 min

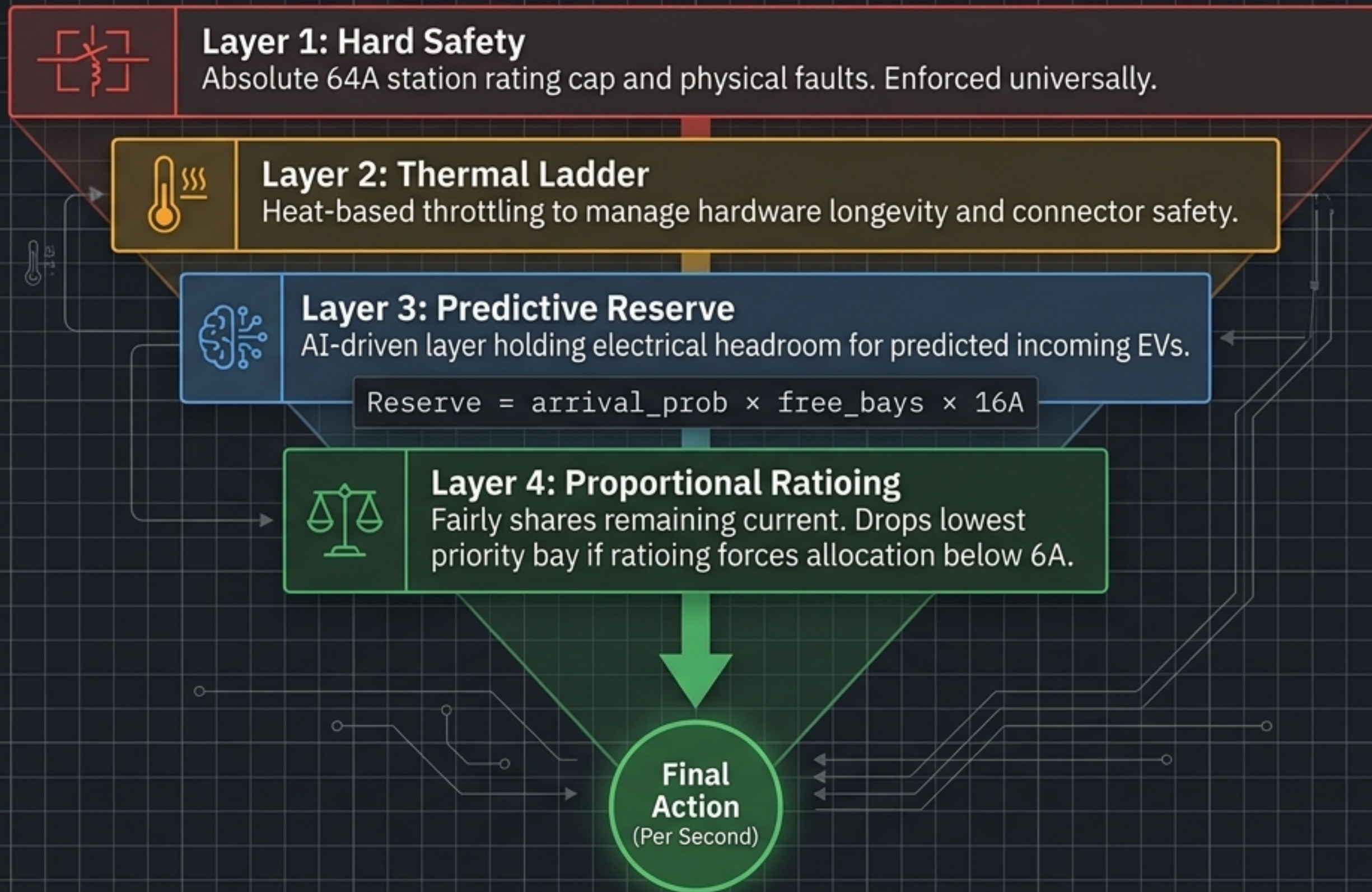
R²: 0.91

DEPLOYMENT PIPELINE: FROM HEAVY ML TO EDGE LOGIC

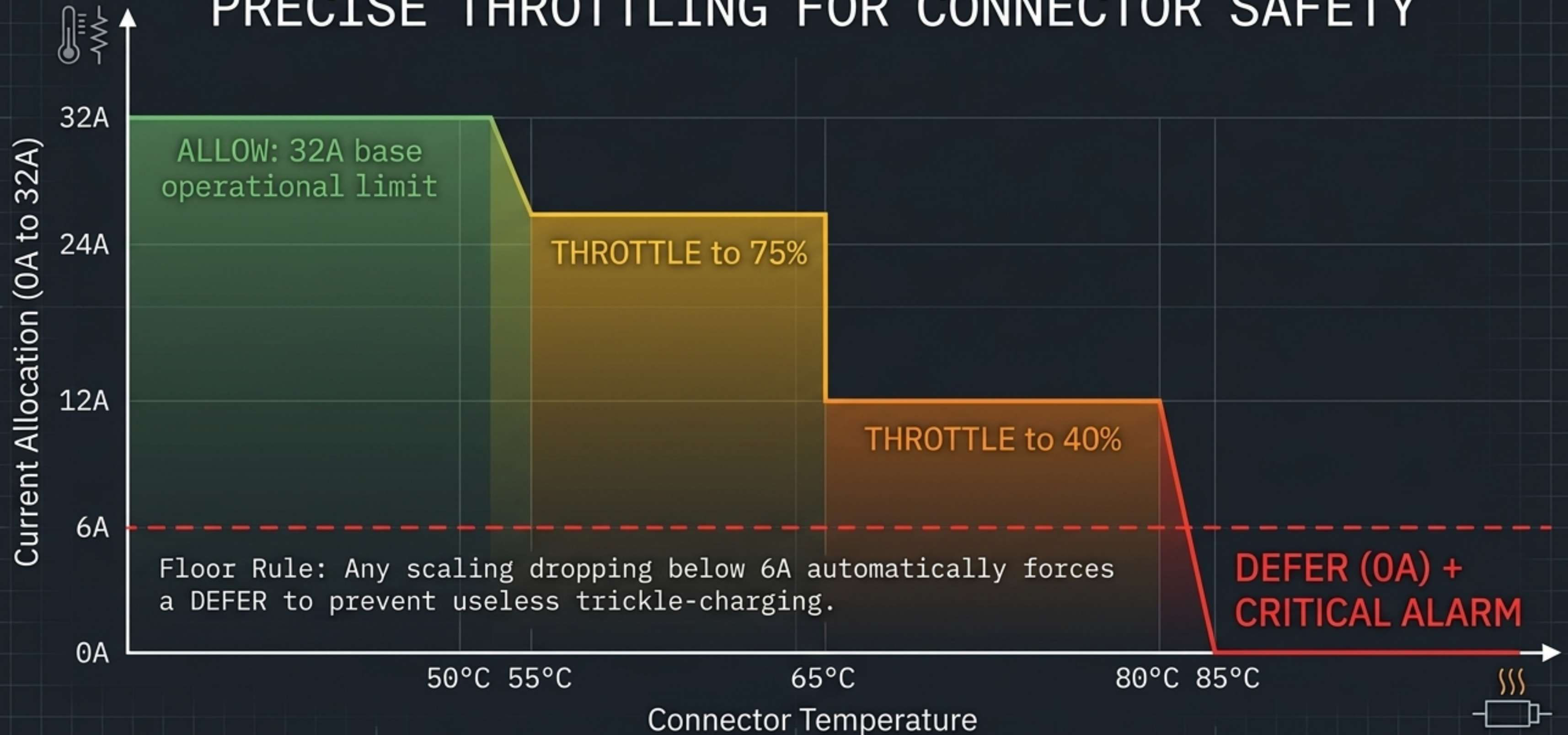


Zero ML libraries on the device. Microsecond execution on a \$10 ESP32.
No TensorFlow Lite. No cloud inference latency.

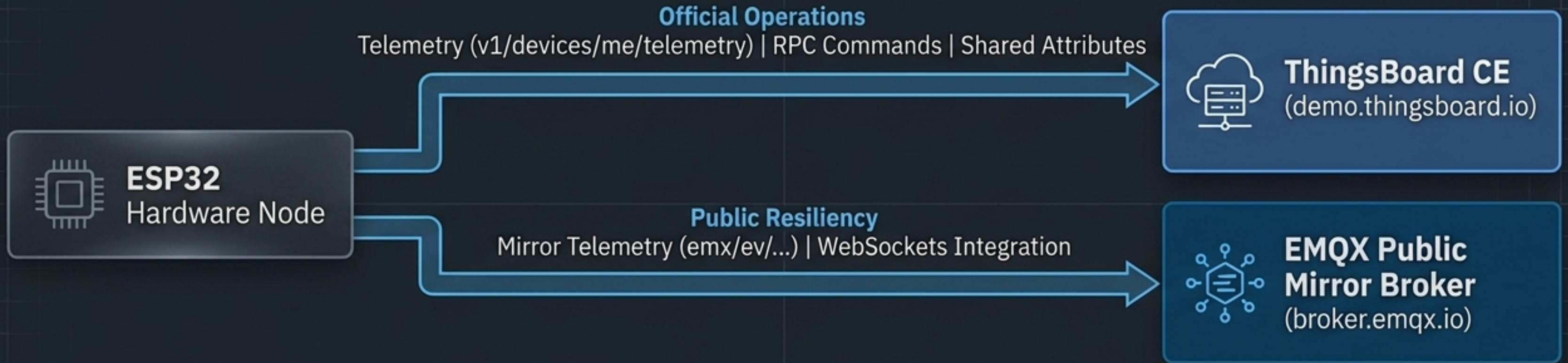
OPTIMIZER PRIORITY: CASCADING FILTER HIERARCHY



THERMAL-CURRENT STEP-DOWN LADDER: PRECISE THROTTLING FOR CONNECTOR SAFETY



DUAL-STREAM CLOUD ARCHITECTURE: RESILIENCY THROUGH DECOUPLING



Strategic Insight Card

Architectural Choice: This dual-stream decouples the front-end dashboard from ThingsBoard authentication, ensuring public, real-time live demonstrations function flawlessly.

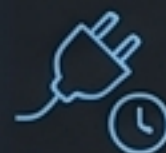
SMART CHARGING STATION: LIVE DASHBOARD INTERFACE

Station Load SVG & AI Arrival Charts



Bay Cards: SoC, Telemetry & Decision Chips

Manual Control Overrides



Offline Fallback Capability (js/demo.js)

If the MQTT connection drops for 12 seconds, the dashboard flawlessly switches to 'DEMO FEED'. A full JavaScript port of the entire station, optimiser, and AI takes over locally. The presentation never fails.

SYSTEM VALIDATION & STANDARDIZATION: PROVEN INTEGRITY & UNIVERSAL DATA CONTRACT

```
[PASS] 14/14 checks passed  
[PASS] Tree export matches sklearn (max err 0.0000 min)  
[PASS] Hard limit respected (worst 64.0A vs limit 64A)  
[PASS] Critical temp defers bay
```

Running regression suite...

Learning export matches and Python simulator

Running regression suite...

Execution time: 4.5s

```
{  
  'station_load_pct': <number>  
  'arrival_probability_pct': <number>  
  'bay1_decision': <'ALLOW' | 'THROTTLE' | 'DEFER'>  
  'alarm_codes': 'TEMP_HIGH | OVERLOAD'  
}
```

Flat-Key JSON Schema for
Cross-Platform Interoperability

Universal Data Contract: One identical schema governs the C++ firmware, the Python simulator, the offline JavaScript engine, and the front-end dashboard.



Sub-second control

Control decisions are made locally every single second without network round-trips.



Tolerant of offline conditions

No critical safety loop relies on an active internet connection.



Zero cloud inference bills

Compute power scales infinitely with physical hardware deployment.



Absolute data privacy

No raw vehicle data leaves the physical site. The cloud is utilised purely for visibility, never for physical control.

Why Edge AI over Cloud AI?